Cambridge (CIE) A Level Maths: Mechanics

Your notes

Quantities, Units & Modelling

Contents

- Scalars & Vectors
- Fundamental Units
- Derived Units
- Types of Force
- Modelling Assumptions

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Your notes

Scalars & vectors

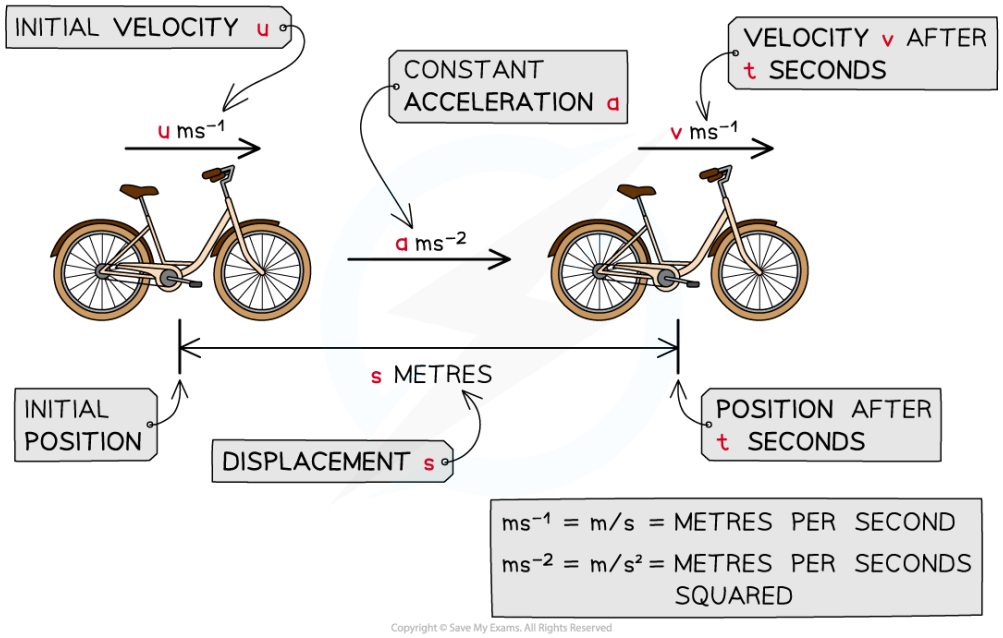
Mechanics uses a variety of different scalar and vector quantities.

What are scalars?

- **Scalars** are quantities without direction, they only have a **size (magnitude)**
  - e.g. a **speed** of 7 m/s, a **distance** of 5 m, a **time** of 15 s or a **mass** of 12 kg
- Scalars can only be **positive**- you can't have a negative distance or speed!

What are vectors?

- **Vectors** are quantities which also have a **direction**, this is what makes them more than just a scalar
  - e.g. two objects with **velocities** of 7 m/s and -7 m/s are travelling at the **same speed** but in **opposite directions**
- Vector quantities can have **positive** or **negative components**
- Some examples of vector quantities you may come across are **displacement**, **velocity**, **acceleration**, **force/weight**, **momentum**
  - **Displacement** is the distance moved in a given direction from a starting point
  - **Velocity** is a speed in a given direction (displacement over time)
  - **Acceleration** is the change in velocity over time



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Worked Example

State whether each of the following is a scalar or a vector quantity.

a) A speed boat travels at 3m/s on a bearing of 052°

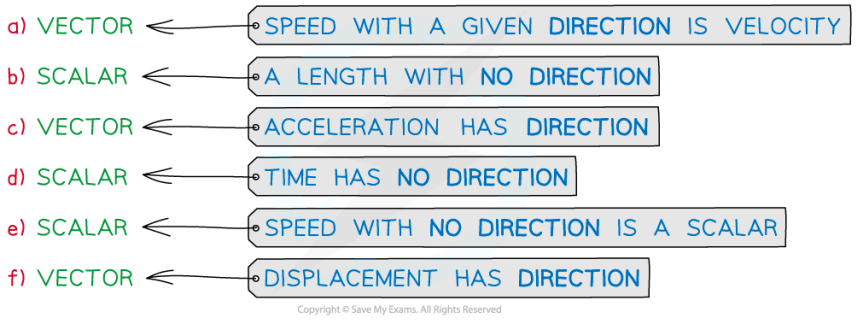
b) A garden is 1.7 m wide

c) A car accelerates at 5.4 ms^{-2}

d) A film lasts 2 hours 17 minutes

e) An athlete runs at an average speed of 10.44 ms^{-2}

f) A ball rolls forwards 60 cm before stopping



Examiner Tips and Tricks

- Make sure you fully understand the definitions of all the words in this section so that you can be clear about what your exam question is asking of you
- Vectors appear in **bold** (non-italic) font in textbooks, on exam papers, etc (i.e. ***F***, ***a***) but in handwriting should be underlined (i.e. $\underline{F}$, $\underline{a}$)

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Fundamental units

What are fundamental units?

- **Fundamental units** or **S.I. units** are the international standardised units used around the world
- The three main units you will be using are:
  - **Length** measured in **metres (m)** (also called displacement)
    - 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm
  - **Time** measured in **seconds (s)**
    - 1 hour = 60 minutes, 1 minute = 60 seconds
  - **Mass** measured in **kilograms (kg)**
    - 1 kg = 1000 g, 1 g = 1000 mg
- All other quantities can be measured in units **derived** from S.I. units

Worked Example

Convert the following into S.I. units:

a) 27.8 km

b) 3.8 hours

c) 0.054 g

a) SI UNIT FOR LENGTH IS METRES (m)

27.8 km × 1000 = 27 800 m

1 km = 1000 m

b) SI UNIT FOR TIME IS SECONDS (s)

3.8 hours × 3600 = 13 680 s

1 HOUR = 60 MINS x 60 SECONDS = 3 600 SECONDS

c) SI UNIT FOR MASS IS KILOGRAMS (kg)

0.054 GRAMS ÷ 1000 = 0.000054 kg = 5.4×10^{-5} kg

1 kg = 1000 GRAMS

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Examiner Tips and Tricks

- Make sure you fully understand the definitions of all the words in this section so that you can be clear about what your exam question is asking of you.
- Make sure you are confident converting between fundamental S.I Units

  e.g. 17 minutes and 42 seconds = 1062 seconds
  2.4 km = 2400 m

  86000 mg = 0.086 kg.

- Watch out for exam questions using non S.I. units such as feet and inches or miles per hour.
- Remember mass (measured in kg) is not a weight (a force measured in Newtons).

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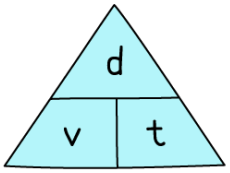
Derived units

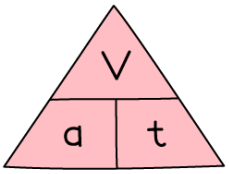
What are derived units?

- Derived units for quantities such as velocity, acceleration and force etc. are combinations of S.I. units
  - **Speed** or **Velocity** = distance ÷ time = **m/s or m s^{-1}** (Velocity is just speed with direction)
  - **Acceleration** = velocity ÷ time = ms^{-1} /s = **m/s^2 or m s^{-2}**
  - **Weight** or **Force** = mass x acceleration = **kg m s^{-2} or N** (N = Newtons)

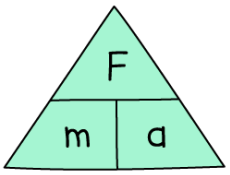
IF YOU EVER FORGET A FORMULA, USE THE UNITS IT IS MEASURED IN TO FIGURE IT OUT
e.g. m / s = METERS ÷ SECONDS = DISTANCE ÷ TIME = VELOCITY



SPEED OR VELOCITY = DISTANCE ÷ TIME
m / s OR ms^{-1}
(VELOCITY IS JUST SPEED WITH DIRECTION)



ACCELERATION = VELOCITY ÷ TIME
m / s^2 OR ms^{-2}



WEIGHT OR FORCE = MASS × ACCELERATION
kgms^{-2} OR N
(N = NEWTONS)

MAKE SURE YOU ARE CONFIDENT CONVERTING BETWEEN THESE DERIVED UNITS e.g. 5.8 kmh^{-1} = 1.6 ms^{-1}
USING 1km = 1000 m AND 1hr = 3600 s
5.8 × 1000 ÷ 3600 = 1.61 CONVERTS kmh^{-1} INTO ms^{-1}

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Worked Example

a) Convert the following into S.I. units

i) 72 km h^{-2} ii) 50 cm per minute iii) 14 g cm^{-3}

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Deal with each conversion in turn

a) (i) Derived SI unit for acceleration is $m\ s^{-2}$

$72\ km\ h^{-2} \times 1000 \div 3600^2 = 0.00556\ m\ s^{-2}$

1 km = 1000 m

1 hour = 3600 seconds

(ii) Derived SI unit for speed/velocity is $m\ s^{-1}$

$50\ cm\ min^{-1} \div 100 \div 60 = 0.00833\ m\ s^{-1}$

1 m = 100 cm

1 min = 60 seconds

(iii) If the units are unfamiliar carry out each conversion separately

$14\ g\ cm^{-3} \div 1000 \times 100^3 = 14000\ kg\ m^{-3}$

1 kg = 1000 g

1 m = 100 cm

Be careful here, this is g/cm^3 so we should multiply by 100^3 instead of dividing

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b) A cyclist takes 15 minutes to travel 2.54 km. Calculate the average speed in $m\ s^{-1}$?

b) It is easiest to convert the units before calculating with them

15 mins = 900 seconds

× 60

1 min = 60 seconds

2.54 km = 2540 m

× 1000

1 km = 1000 metres

Remember SI units are:
Metres (m)
Seconds (s)
Kilograms (kg)

$$S = \frac{D}{T} = \frac{2540}{900} = 2.82\ m\ s^{-1}$$

If you do use original units to calculate make sure you convert the units at the end

$$\frac{2.54}{15} \times 1000 \div 60 = 2.82\ m\ s^{-1}$$

1 km = 1000 m

1 min = 60 seconds

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Examiner Tips and Tricks

- Watch out for exam questions using non S.I. units such as feet and inches or miles per hour.
- Make sure you are confident **converting** between these derived units,
- Remember **speed** is velocity without direction and **weight** is a force measured in **Newtons**, not a mass measured in kg.

- If you ever forget a formula, use the units it is measured in to figure it out,
- e.g. m/s = metres ÷ seconds = distance ÷ time = velocity.

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Types of Force

What is a force?

- A force is a **vector** quantity, it has both **magnitude** and **direction**
- A force is a push or a pull on an object
- A force is measured in **Newtons**
- In A level mathematical models, forces act at a **single point** called a **particle** which occupies a single point in space

What types of force are used in mechanics?

- **Weight** is the effect of **mass** and **gravity,** it always acts downwards
- **Tension** is a pulling force, it always acts away from an object
- **Thrust** is a pushing force, it always acts towards an object
- **Friction** is a resistive force, it acts to oppose the motion of an object
- Every surface will produce a **reaction force**, it will always act **perpendicular** to the surface

Mass, Gravity and Weight

- **Mass** is a measure of the amount of matter in an object
  - **Mass** is a **scalar** quantity, measured in **kilograms**, **kg**
  - **Mass** is universal, it does not change based on location
- **Gravity** is the force by which a body (usually a planet or a star) pulls objects towards its centre
  - **g** is the **acceleration** due to **gravity, measured in m s^{-2}**
  - On Earth, **g** is approximately **10 m s^{-2}** , although its exact value varies with location
  - **g** is different elsewhere in the universe
- **Weight** is the product of **mass** and **gravity**, it is a **force** measured in **Newtons**
  - **Weight** is a **vector** quantity
  - **Weight** varies with location
  - **Weight** always acts vertically downwards towards the ground

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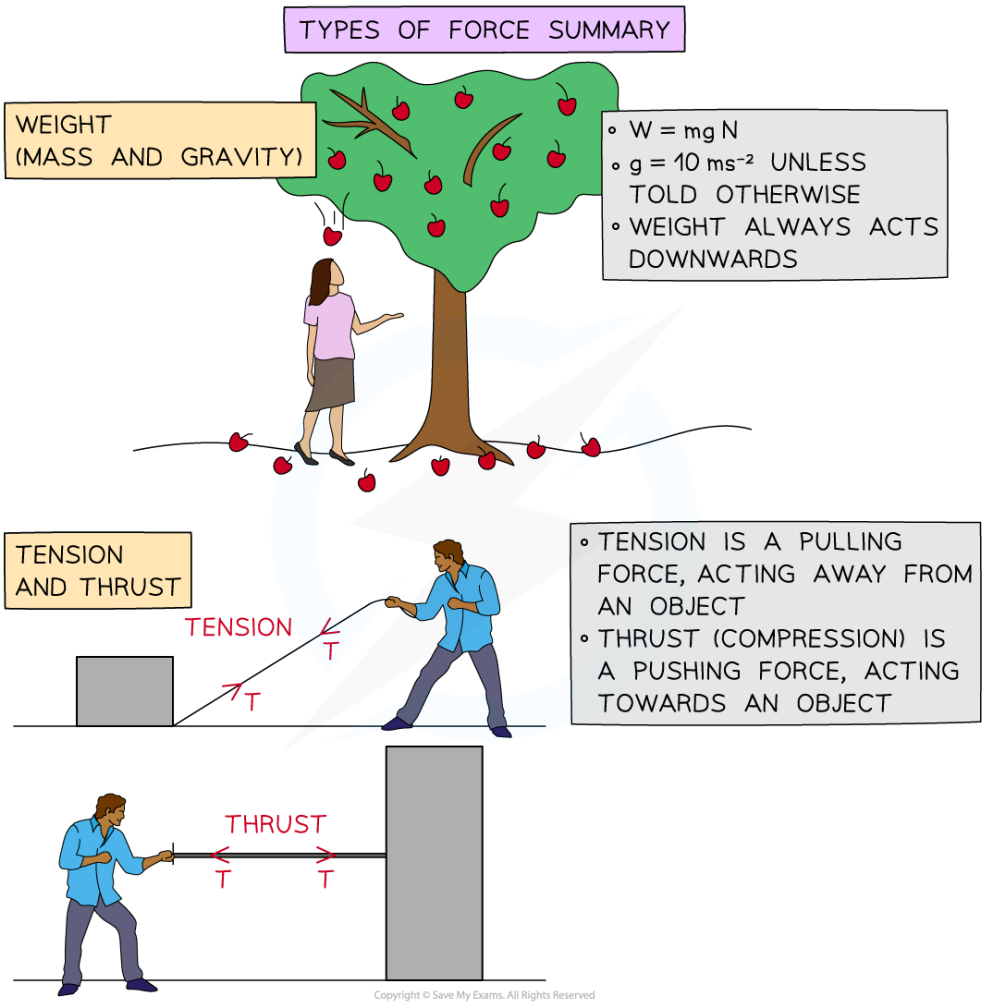
TYPES OF FORCE SUMMARY
WEIGHT (MASS AND GRAVITY)
W = mg N
g = 10 ms⁻² UNLESS TOLD OTHERWISE
WEIGHT ALWAYS ACTS DOWNWARDS
TENSION AND THRUST
TENSION
T
T
TENSION IS A PULLING FORCE, ACTING AWAY FROM AN OBJECT
THRUST (COMPRESSION) IS A PUSHING FORCE, ACTING TOWARDS AN OBJECT
THRUST
T
T
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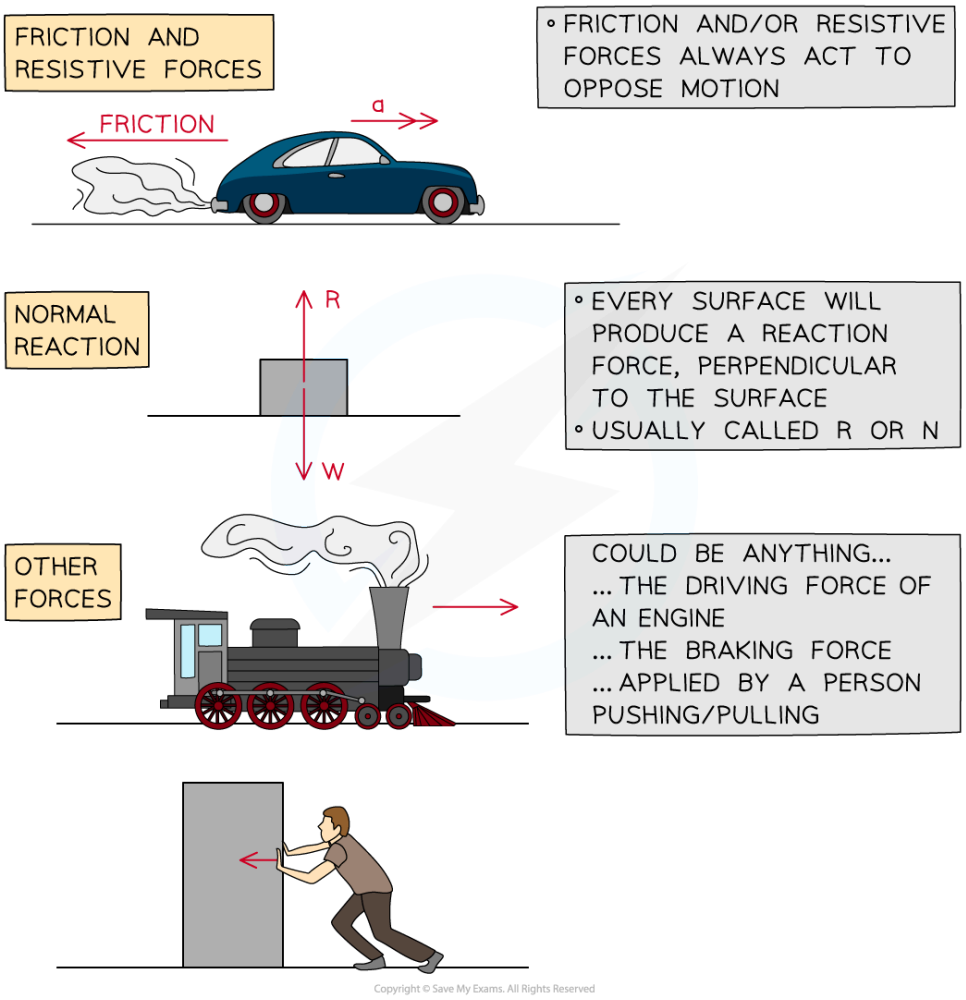
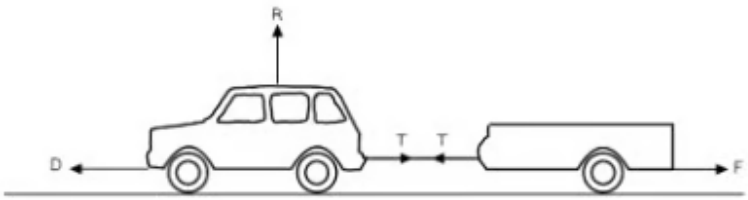


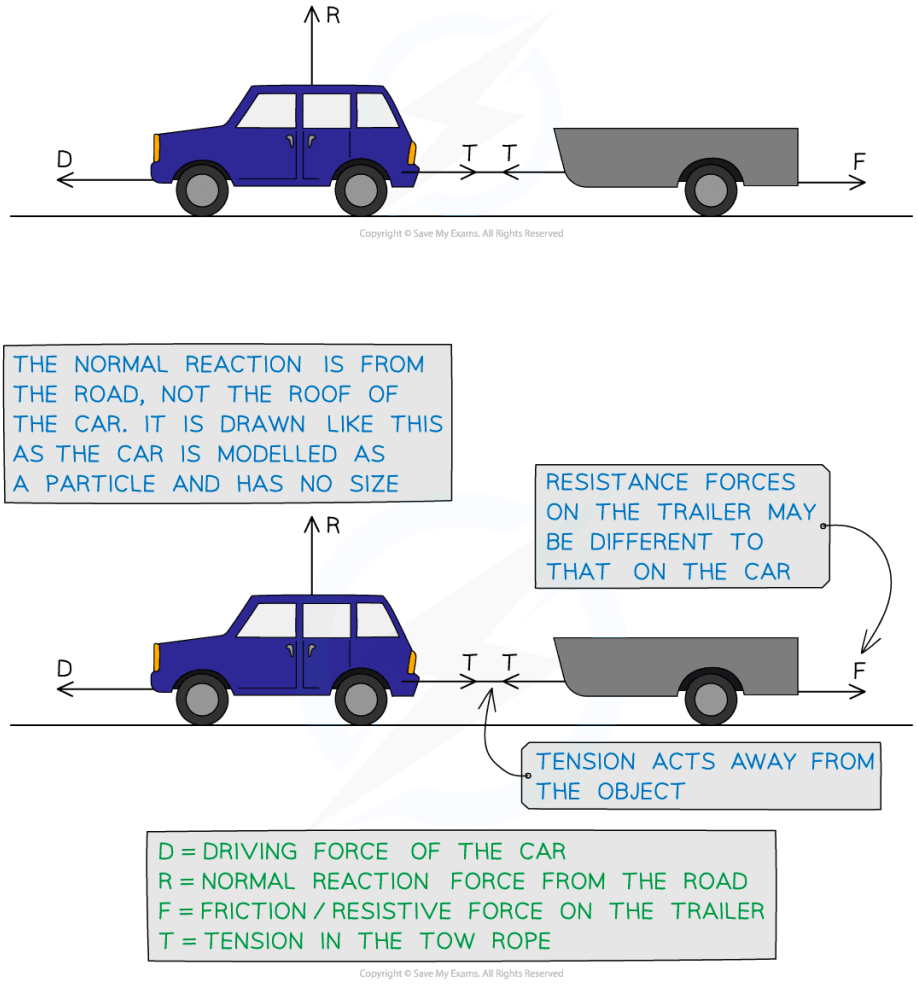
Worked Example

The diagram below shows a car towing a trailer by a light inextensible string.

Forces *D*, *R*, *F* and *T* N act on the car and caravan as shown.

Write a brief description of which each force represents.





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Examiner Tips and Tricks

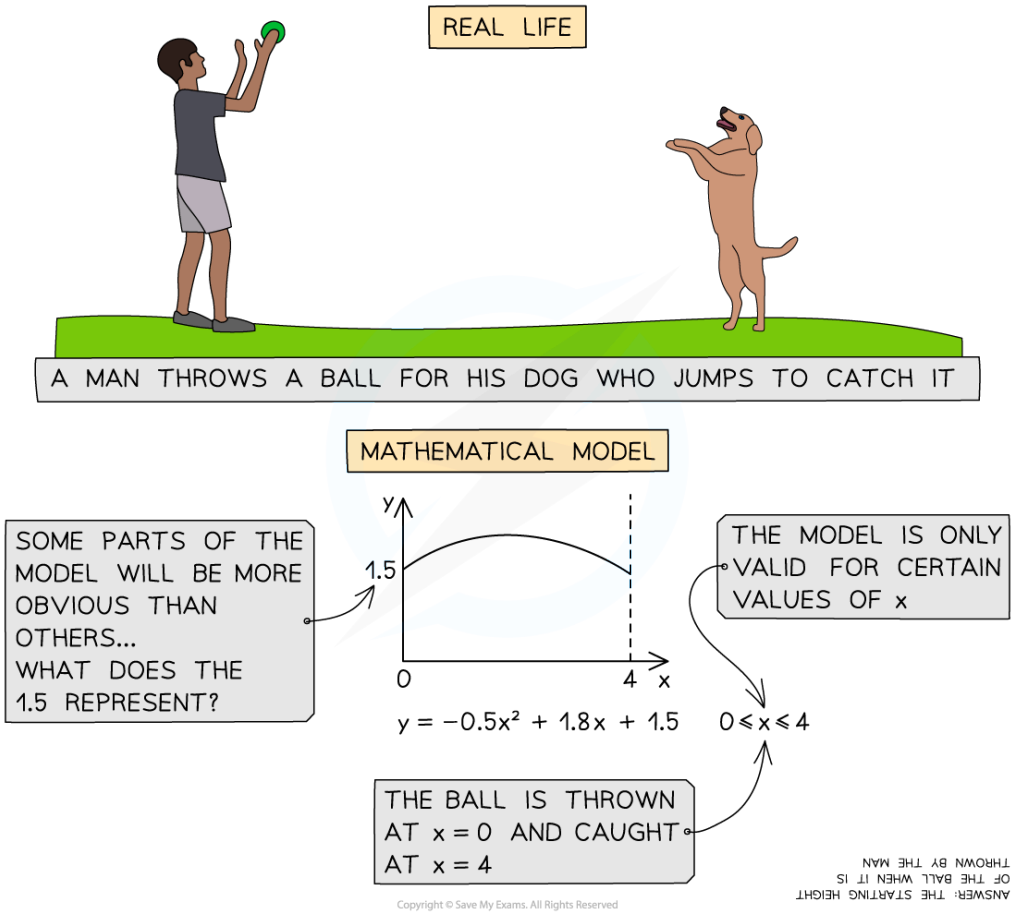
- Always sketch a diagram and mark the forces on the diagram clearly, it will help you to understand the problem. Add more things to the diagram as you progress through the question. You may not even need all the forces from your diagram but it is useful to have them there.
- You should always round your answers to three significant figures.

Your notes

Modelling Assumptions

What is modelling in Mechanics?

- Mechanics uses **modelling** to solve problems in real life situations
- We use **assumptions** to simplify real life problems in order to turn them into equations or graphs that can be solved
- We will sometimes need to **criticise** or **refine our assumptions** to improve the model



Modelling Assumptions in Mechanics:

- There are many different **modelling assumptions** that can be made when dealing with **real life situations**
- Different assumptions will be needed for different models
- The assumptions you make will affect the calculations in each problem differently

What types of modelling assumptions are there?

- **Gravity** is constant and vertical
- **Air resistance** is usually modelled as **negligible** and can be ignored
- A **smooth** surface has no **friction**
- A **rough** surface has a **frictional force** between the surface and any object that makes contact with it
- A **particle** has **negligible** dimensions, therefore forces will all act on a particle at the same point
- A **rod** or a **beam** should be treated as a long, rigid particle
- A **uniform** object's **mass** is distributed evenly
- A **light** object has zero **mass**
- An **inextensible** object cannot be stretched

Your notes

Worked Example

Two blocks, A and B are attached by means of a light inextensible string running over a smooth pulley. Block A has mass 0.7 kg and is accelerating along a smooth horizontal surface, block B has a mass ().2 kg. Both A and B are modelled as particles. State how the following modelling assumptions can be used in your calculations:

a) A and B are both particles.

b) The string is light.

c) The string is inextensible.

d) The pulley is smooth.

e) The surface A is moving along is smooth.

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a) Dimensions and air resistance can be ignored

Particles have no dimensions so air resistance cannot act on the object

b) The tension is constant throughout the string (same at A and B)

Light objects have zero mass

c) Acceleration will be the same for both blocks

Inextensible means the string cannot stretch

d) The tension in the string is the same on both sides of the pulley

Smooth surfaces provide no friction

e) There is no frictional force between the surface and block A

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